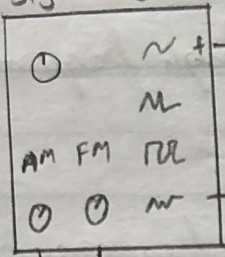
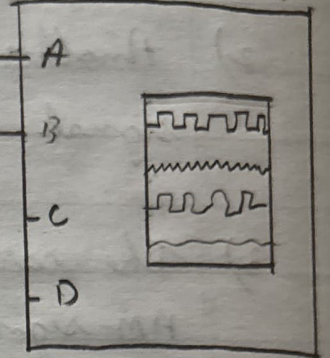


DSO Input

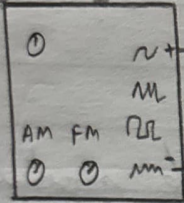
Signal generator



Oscilloscope



Signal generator



ground

Fig 2.1

Practical - 2AIM:-

To generate and observe frequency modulation (FM) using generator & DSO.

Objective:-

To generate a Frequency Modulated (FM) signal using a generator & to observe the waveform of FM on a Digital Oscilloscope (DSO). The objective also includes studying how the frequency of the carrier varies with the modulating signal while the amplitude remains constant.

Theory:-

Frequency Modulation (FM):

Frequency Modulation is a Modulation technique in which the frequency of a high frequency carrier signal is varied accordance with the instantaneous amplitude of a low frequency modulating (message) signal, while amplitude of carrier remains constant.

Working Principle:-

- 1) One signal generator produces a high frequency carrier signal.
- 2) Another signal produces a low frequency modulating signal.
- 3) The modulating signal controls the frequency deviation of the carrier.
- 4) The resulting FM signal is applied to the DSO.
- 5) The DSO displays constant amplitude carrier, variable frequency corresponding to the modulating signal. From the waveform, the carrier cycles become compressed and expanded, indicating frequency variation.

Amplitude

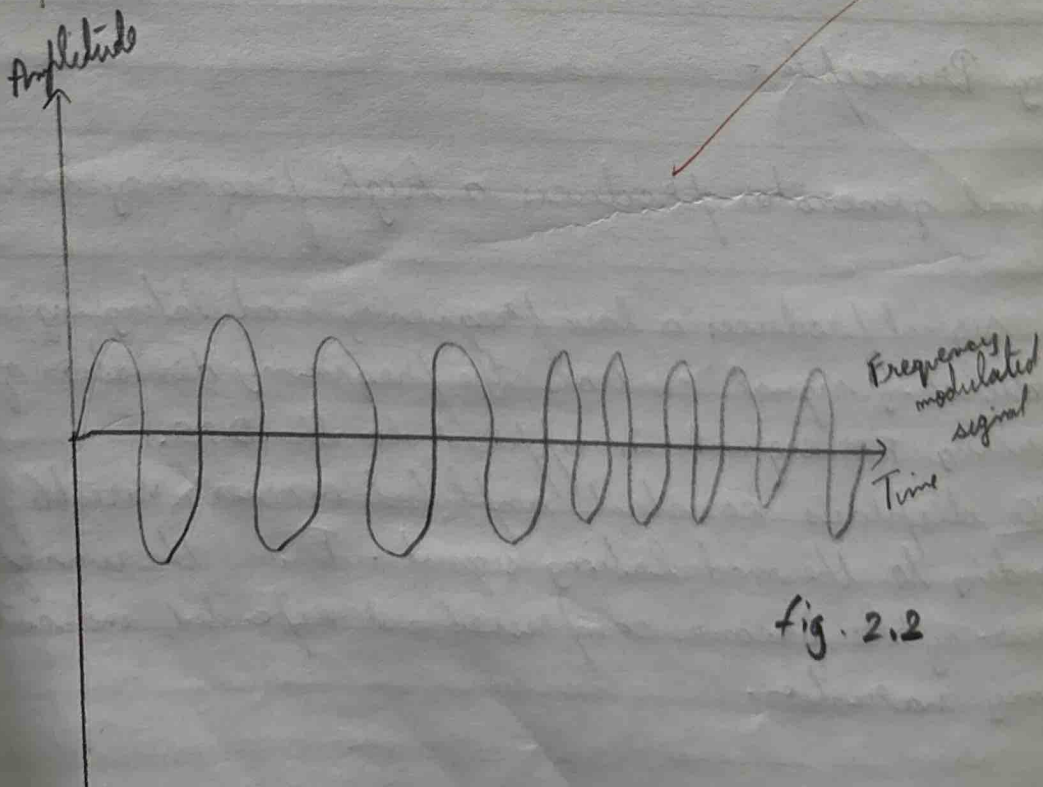
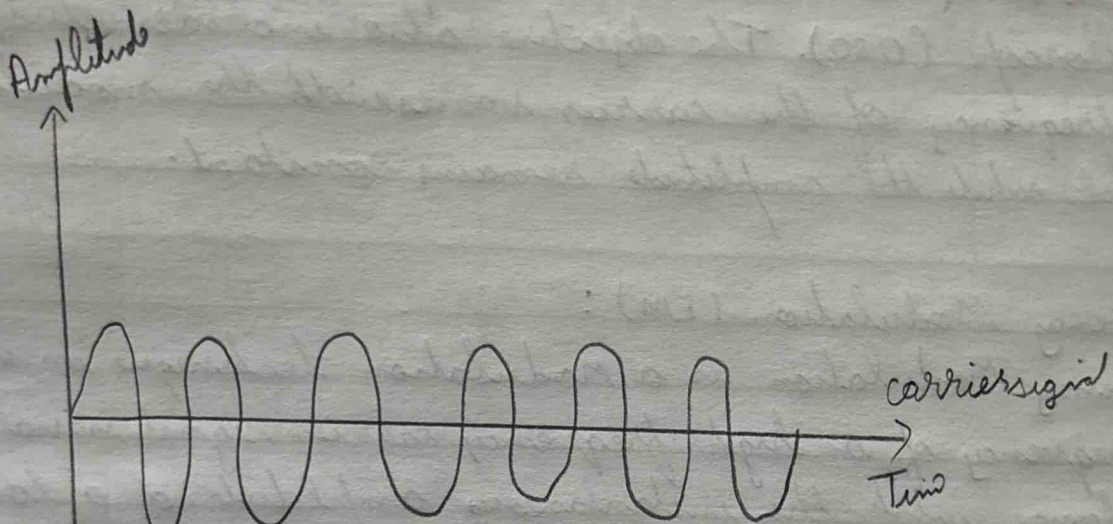
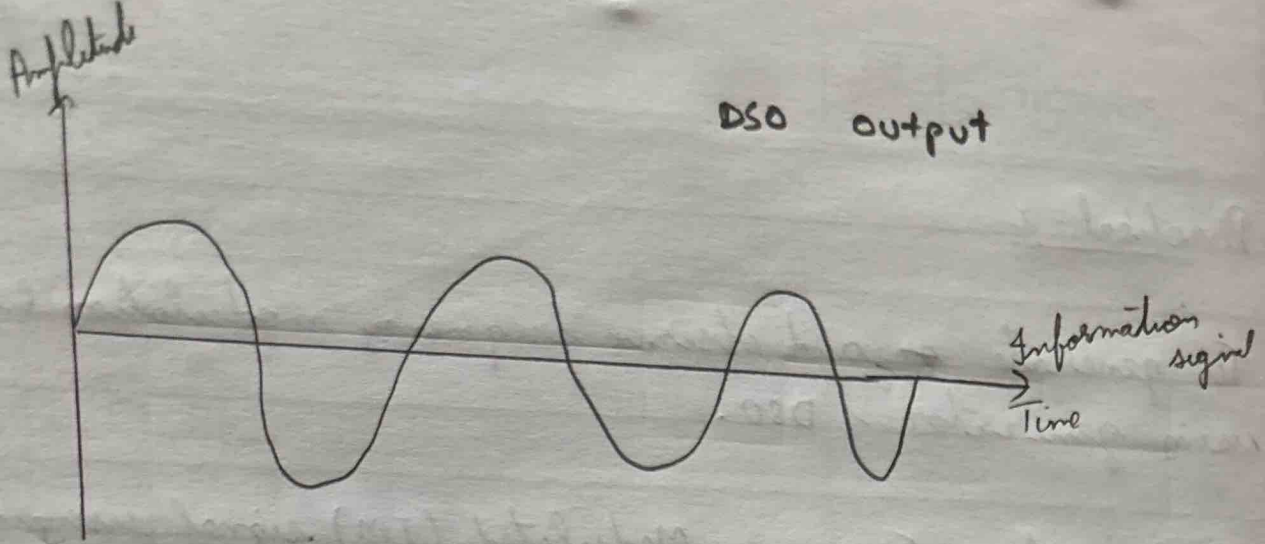


fig. 2.2

Observation :

- The amplitude of the FM signal remains constant throughout.
- The frequency of the carrier varies according to the modulating signal.
- Regions of higher modulating signal amplitude produce higher frequency deviation.
- The DSO clearly displays frequency variation through changing waveform density.

Conclusions :

- Frequency Modulation was successfully generated using a signal generator and observed on a digital storage ~~oscilloscope~~ oscilloscope. The experiment confirms that in FM, the carrier frequency varies in accordance with the modulating signal while the amplitude remains constant. ~~The~~ The variation in spacing between waveform cycles on the DSO verifies the principle of frequency modulation. This experiment demonstrates the ~~effectiveness~~ effectiveness of FM in reducing noise & its importance in modern communication system.

Plu
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